

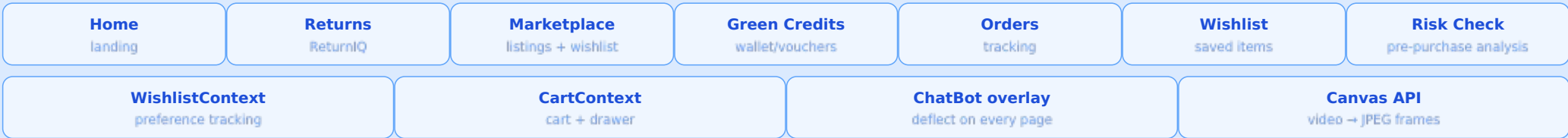
Second Life Commerce — Technical Architecture

HackOn with Amazon Season 6.0 · AI-Powered Returns & Sustainable Resale

CLIENT

React 18 + Vite Frontend

Deployed on Vercel · /api/* proxied to Render backend · Tailwind CSS

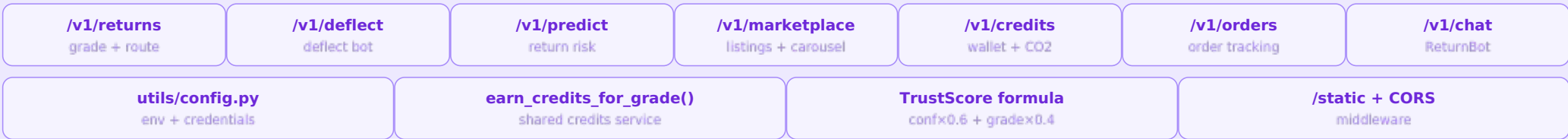


HTTP /api/*

API LAYER

FastAPI Backend (Python)

Deployed on Render · SQLAlchemy ORM · auto-seeds 30 demo scenarios



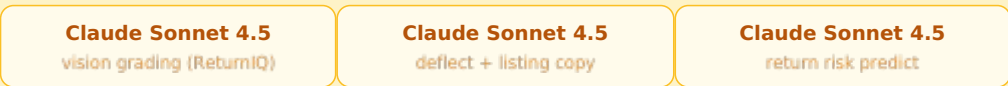
primary ↓

fallback ↓

AI SERVICES

AWS Bedrock (primary)

auto-scales · multi-region · Claude Sonnet 4.5



Groq (automatic fallback)

zero-downtime hot failover · no code changes needed



if Bedrock unavailable →



DATA LAYER

Data Layer

SQLAlchemy ORM abstracts DB · swap SQLite → DynamoDB or Aurora Serverless with one config change

